

Impact of Shared E-scooter Introduction on Public Transport Demand: A Case Study in Santiago, Chile

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Abstract

This study examines how the introduction of shared electric scooters (e-scooters) affects public transport demand in Santiago, Chile, analyzing whether they complement or substitute for existing transit services. We used smart card data from the integrated public transport system of Santiago and GPS traces from e-scooter trips during the initial deployment period. We employed a difference-in-differences approach with negative binomial regression models across three urban regions identified through k-means clustering: Central, Intermediate, and Peripheral. Results reveal spatially heterogeneous effects on public transport boardings and alightings. In the Central Region, e-scooter introduction was associated with significant substitution

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effects, showing a 23.87% reduction in combined bus and metro boardings, suggesting e-scooters replace short public transport trips in high-density areas. The Intermediate Region showed strong complementary effects, with a 33.64% increase in public transport boardings and 4.08% increase in alightings, indicating e-scooters successfully serve as first/last-mile connectors that enhance transit accessibility. The Peripheral Region exhibited no significant effects. Metro services experienced stronger impacts than bus services, with metro boardings increasing 9.77% in the Intermediate Region. Our findings advance understanding of micromobility-transit interactions by demonstrating that both substitution and complementarity can coexist within the same urban system, depending on local accessibility conditions. These results highlight the need for spatially differentiated mobility policies that recognize e-scooters' variable roles across urban environments.

Keywords: Electric scooters, e-scooters, Micromobility, Mobility as a Service, Public Transport, Public Transit

1. Introduction

Cities worldwide need to understand how shared electric scooters (e-scooters) affect public transport as they invest in sustainable mobility solutions. In dense urban areas, micromobility (the use of small, lightweight vehicles weighing less than 500 kg (Dediu, 2019)) can reduce traffic, cut emissions, and improve local economies (Tuli et al., 2021). However, the impact of e-scooters on transit ridership remains unclear. Previous studies show mixed

effects: some report reduced public transport demand (Vinagre Díaz et al., 2023), while others find e-scooters acting simultaneously as competitors and complements to transit (Yan et al., 2021; Aarhaug et al., 2023).

Shared e-scooters have emerged as a prominent micromobility option through Mobility as a Service (MaaS) applications (Narayanan and Antoniou, 2023). These services let users locate, reserve and ride scooters via mobile phones, with flexible parking options. Private companies typically operate these services and market them as complementary to public transit, particularly for first/last mile connections (Zuniga-Garcia et al., 2022). While they could enable modal shifts and complement public transit (Latinopoulos et al., 2021; Zuniga-Garcia et al., 2022), they also raise concerns about vehicle displacement, unauthorized space use, safety (Stigson et al., 2021; Graells-Garrido et al., 2025), pedestrian conflicts (Gössling, 2020), and environmental impacts (Echeverría-Su et al., 2023).

Current knowledge gaps are significant. Most research analyzes e-scooter effects in specific city areas or transit lines, but no study examines how effects vary across different urban structures such as central business districts versus peripheral zones. Additionally, most studies focus on single public transport modes (either bus or metro) rather than evaluating effects on integrated multimodal systems. This represents an important limitation because e-scooters may impact central areas differently than suburban areas, and may affect bus services differently than metro services.

Our study examines three critical research questions: *How does introduc-*

ing shared e-scooters affect public transport ridership across different urban contexts? Do shared e-scooters effectively solve first/last mile challenges for public transport? How do these effects vary by transport mode (bus vs. metro) and urban region? The answers can help urban planners and transit agencies make informed decisions about e-scooter regulations, infrastructure planning, and integration with existing transit systems.

Santiago, Chile provides an ideal case study. The city has seen significant transport changes, including reduced transit use and increased motorized transport (Graells-Garrido et al., 2023a), while introducing e-scooters during our study period. Most research examines U.S. and European cities, leaving gaps in understanding how e-scooters affect transit in other contexts. This gap is especially relevant in South America (Mitropoulos et al., 2023), where cities like Santiago face different urban dynamics and transit challenges than well-studied North American and European contexts.

Using a difference-in-differences methodology with comprehensive smart card and GPS data, we advance the field in three ways. First, we demonstrate that both substitution and complementarity coexist spatially within the same urban system, with substitution dominating in transit-rich central areas (-23.9% in boardings) and complementarity emerging in intermediate zones (+33.6% in boardings). This spatial heterogeneity explains why previous studies have reached different conclusions across various urban contexts. Second, we contribute methodologically by developing an automated clustering approach using k-means regionalization that eliminates arbitrary manual

selection of treatment and control zones common in prior quasi-experimental studies, enhancing the validity of causal inference. Third, we provide the first systematic evidence of mode-specific effects (bus vs. metro) in a Latin American context, revealing that metro services show stronger complementary relationships with e-scooters than bus services.

The paper proceeds as follows: Section 2 reviews literature on e-scooter impacts. Section 3 describes our data and study context. Section 4 details our methodology, including clustering and regression approaches. Section 5 presents results, followed by discussion in Section 6. Section 7 concludes with implications for urban mobility strategies.

2. Literature Review

Research on e-scooter impacts employs two primary methodological approaches: spatial analysis and quasi-experimental methods. Each approach reveals different aspects of the e-scooter-transit relationship, but significant gaps remain in understanding spatial heterogeneity and multimodal effects.

Spatial Analysis Studies

Spatial analysis studies classify e-scooter trips as complementary or substitutive based on proximity to transit stations. These studies consistently find mixed effects within the same urban areas. Research in Rome found 50% of e-scooter trips either replace public transport or serve areas with inadequate transit infrastructure (Vinagre Díaz et al., 2023). Analysis in

Washington, DC showed approximately 70% of trips substituting for transit while 20% complemented it (Yan et al., 2021). Studies in Norwegian cities demonstrated that e-scooters simultaneously compete and complement transit (Aarhaug et al., 2023). Research in Austin and Minneapolis found 47–52% of trips competing with transit and 29–31% complementing it (Bai and Jiao, 2020). Additional studies in Austin showed increased e-scooter use near transit stations (Jiao and Bai, 2020), while Louisville research found positive correlation between e-scooter usage and bus stop density, suggesting complementary effects (Hosseinzadeh et al., 2021). A different methodological approach using stated preference surveys in Seoul found that 40.6% of travelers preferred e-scooters, 39.2% chose buses, and 20.2% opted for walking (Baek et al., 2021). Additional studies in Austin showed increased e-scooter use near transit stations (Jiao and Bai, 2020).

These mixed findings suggest that spatial context mediates e-scooter-transit relationships, yet most studies aggregate effects across entire urban areas without examining how local characteristics influence outcomes.

Quasi-Experimental Studies

Quasi-experimental research provides stronger causal evidence but focuses narrowly on specific modes or routes. Bus-focused studies show varying results: research in Indianapolis found 27% of e-scooter trips competed with buses while 29% increased bus usage (Luo et al., 2021). Studies in Louisville showed complementary effects on express routes (Ziedan et al., 2021a), while

analysis in Nashville found different effects by trip purpose (Ziedan et al., 2021b).

Rail-focused studies consistently show complementary effects. Research in Seattle found new rail stations significantly increased nearby e-scooter presence (Tyndall, 2022). Studies in Berlin showed e-scooter demand rising near new stations and falling near closed ones (Weschke, 2023).

Research Gaps

Current literature exhibits three critical limitations. First, studies examine either spatial patterns or causal effects, but not both within a comprehensive framework that accounts for urban heterogeneity. Second, most research focuses on single transport modes, missing interactions within integrated multimodal networks. Third, research concentrates on North American and European cities, leaving substantial gaps in understanding different urban contexts, particularly in Latin America where transit systems and urban development patterns differ significantly. Our study addresses these gaps by combining quasi-experimental methodology with systematic spatial analysis across an integrated multimodal network in Santiago, Chile. Unlike previous work that manually selects treatment and control zones, we employ automated clustering to enhance methodological rigor while examining both bus and metro interactions across different urban contexts within a single analytical framework.

3. Data Description

This study examines how e-scooters affect public transport demand in Santiago, Chile’s capital and largest city. As of the 2017 Census, Santiago had a population of nearly 8 million across 867.75 square kilometers, comprising over 40 independent municipalities, with 35 in the urban core and the rest on the periphery (see Fig. 1 a).

We analyze two primary data sources across specific time periods. First, we examined *bip!* smart card records from the second week of May 2018 and 2019, capturing typical urban activity without special events. The 2019 period includes the newly opened Metro Line 3. Second, we analyzed GPS traces from shared e-scooters between October 10th, 2018 and April 15th, 2019, covering the introduction and establishment of these services in Santiago. The dataset includes dockless e-scooters that users could park freely within service areas.

Our analysis uses geographical zones defined by the Metropolitan Public Transport Directory (DTPM), comprising 804 transportation analysis zones averaging 1.23 squared km each (see Fig. 1 b).

Santiago’s public transport system integrates metro, urban buses, and rail with a nearly flat fare structure. The system allows up to three transfers within two hours using the *bip!* smart card (the only accepted payment method). The card records trip origins but not destinations, as validation occurs only when boarding buses or entering metro stations. Beyond public transport, the city offers taxi services, ride-hailing, and bike-sharing.



Figure 1: Area under analysis. Left (a): Administrative boundaries of municipalities (orange) and of Traffic Analysis Zones (purple). Right (b): Metro and train lines operating in the city. The red line (Metro Line 3) was inaugurated in late 2018, between our periods under study. OpenStreetMap data and Map tiles © CARTO are used under a Creative Commons License.

The public transport data comes from DTPM’s ADATRAP system (Gschwendner et al., 2016), which consolidates smart card transactions, bus GPS records, and operator information. ADATRAP processes these data using travel pattern analysis and spatial distribution of boarding points (Munizaga and Palma, 2012). We can aggregate trips from bus stops or metro stations into zones (Fig. 2.a shows trip distribution), municipalities, or macro-areas.

Through an agreement with Las Condes municipality (Fig. 4), we obtained GPS traces of 45K e-scooter trips. Using a stop-based algorithm (Graells-Garrido and Saez-Trumper, 2016), we identified trips and excluded staff relocation movements. The data reveals distinct patterns from public transport, as shown in Fig. 3. E-scooter usage shows less pronounced morn-

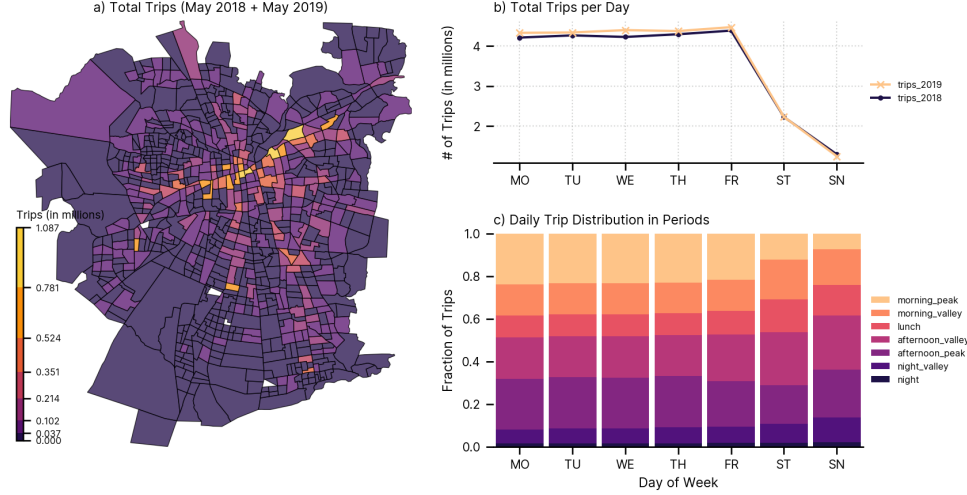


Figure 2: Distributions of Public Transport trips. Left (a): a choropleth map of total trips with respect to Traffic Analysis Zones. Top right (b): total trips per day, per year. Bottom right (c): relative distribution of trip starting periods for each day, where the periods are defined in Table A.8.

ing peaks from 6:00–9:00, while maintaining similar afternoon peaks between 17:30–20:30. We observed higher lunchtime usage between 12:00–14:00, likely due to Las Condes’ concentration of business areas. Shared e-scooter trips are notably shorter, averaging 1.44 km compared to 9.25 km for public transport, with origins and destinations clustering near metro stations.

We complemented these transport datasets with demographic data from the 2017 Chilean Census (National Institute of Statistics, 2017), including population counts by age and sex, as well as average educational level measured in years of schooling. We used educational level as a proxy for income, an approach supported by research linking education to income distribution (Gregorio and Lee, 2002; Jerrim and Macmillan, 2015). Additionally, we

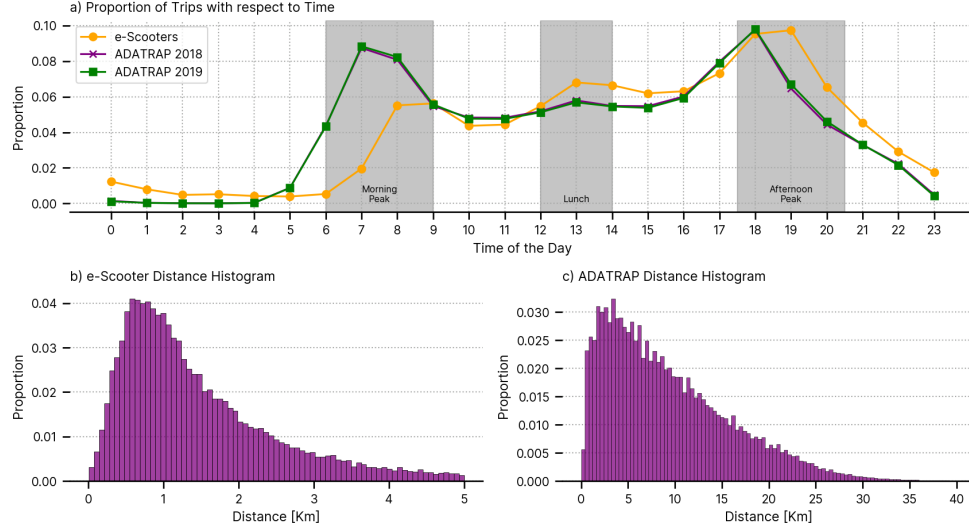


Figure 3: Distributions of trip features from Public Transport and shared e-scooters. Top (a): Proportion of trips with respect to hour of the day, per year. Bottom left (b): Histogram of trip distances in e-scooter trips. Bottom right (c): Histogram of trip distances in Public Transport.

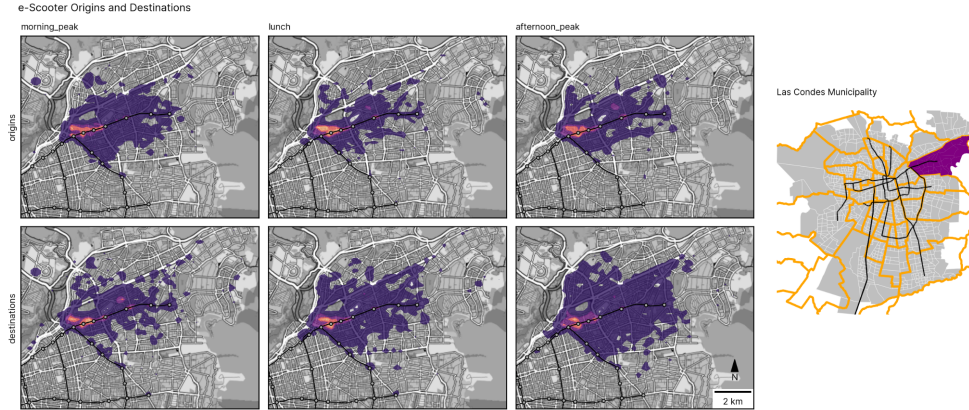


Figure 4: Origins and destinations of shared e-scooter trips. The top row shows trip origins in three periods of the day (morning peak, lunch and afternoon peak), whereas the bottom row shows trip destinations for the same period. The map on the right highlights the area of availability of the shared e-scooter service: the municipality of Las Condes.

used DTPM’s General Transit Feed Specification (GTFS) data to calculate accessibility indicators, which we detail in the methodology section as they relate directly to measuring shared e-scooter effects on transport demand.

4. Methodology

To examine how e-scooters affect public transport use, we conducted a difference-in-differences analysis using geographical zones defined by DTPM as our spatial units. We divided urban areas into three groups using clustering to account for spatial differences and ensure valid comparisons between control and treatment zones. This section details our variable selection and methodology.

Selection of Study Variables

We compiled an initial list of variables that are typically considered to influence travel behavior and assessed their relevance in the context of Santiago and this study. The selected variables include sociodemographic variables from the 2017 census, such as *underage population*, *working population*, *retired population*, *percentage of female population*, *average educational level*, and *population density*. Additionally, we included variables from ADATRAP dataset to capture travel behavior, such as the *average number of public transport trips per person*, *average time of the first public transport trip* made each day, and *average public transport trip distance*. We also included variables to determine each zone’s accessibility to public transport, mapping the metro

network within Santiago and counting the *number of metro stations* and *bus stops per zone* using data from OpenStreetMap and the General Transit Feed Specification (GTFS). These indicators were calculated for each zone in our design. A comprehensive list of these variables can be found in Table 1 below. Additionally, Figure 5 and Figure 6 show the distribution of these variables across the different zones of the city. We can see that there's a central point in the city that is highly densely populated, and as we get farther away from it the population spreads out.

Name	description	source	use	unit	mean	std
under age population	Percentage of the population that's younger than 18.	1	Zone clustering	%	22.50	4.78
working population	Percentage of the population that's between the ages of 18 and 65.	1	Zone clustering	%	65.60	4.44
retired population	Percentage of the population that's older than 65.	1	Zone clustering	%	11.90	4.61
female population	Percentage of the population that's female.	1	Zone clustering	%	51.50	2.11
avg. educational level	Average of years of education received by people over 18.	1	Zone clustering	years	11.86	1.80
bus stops	Number of bus stops in the zone.	2	Zone clustering	count	14.09	8.11
bus stop density	Bus stops per square meter.	2	Zone clustering	stops/km ²	14.15	7.90
bus stops per 1000 people	Bus stops per 1000 inhabitants.	1, 2	Zone clustering	stops/person	2.50	4.13
population density	Inhabitants per square meter.	1	Zone clustering	person/km ²	7696.17	4591.17
avg. number of trips per person	Average number of public transport trips each person makes.	3	Zone clustering Regression	count	1.91	0.28
avg. time of first trip	Average hour of the first trip made on public transport each day.	3	Zone clustering	hour	10.99	0.40
avg. trip distance	Average distance of trips made on public transport originating in the zone.	3	Zone clustering	meters	9218.33	3053.81

Table 1: Summary of Zone Clustering Variables. Each column includes the variable name, a brief description, the data source (represented by a number), the role of the variable in the analysis (use), its unit of measurement, and key statistical indicators such as the mean and standard deviation. The variables span demographic information (e.g., underage population, working population), educational level, and transport-related metrics (e.g., number of bus stops, bus stop density). The data sources are as follows: (1) 2017 Census, (2) OpenStreetMap, and (3) ADATRAP.

Clustering and Definition of Treatment and Control Groups

In this study, we aimed to create meaningful spatial clusters that reflect both statistical similarities and geographic proximity. The process of creating regions, known as regionalization (Duque et al., 2007), is crucial in this context. Regionalization is a specific form of clustering aimed at grouping observations that are alike in both their statistical characteristics and spatial proximity. This ensures that the resulting zones are not isolated but are similar to their neighboring areas, aligning with our goal of meaningful spatial clustering.

We used Moran’s I to analyze spatial autocorrelation and applied two selection criteria: (1) minimum Moran’s I value of 0.25 to ensure meaningful spatial clustering or dispersion, and (2) maximum p-value of 0.05 for statistical significance of the spatial pattern. This ensured that selected variables exhibited statistically significant and substantial spatial dependencies rather than random spatial distribution.

Based on these criteria, we selected the following cluster variables: *percentage of underage population, percentage of working population, percentage*

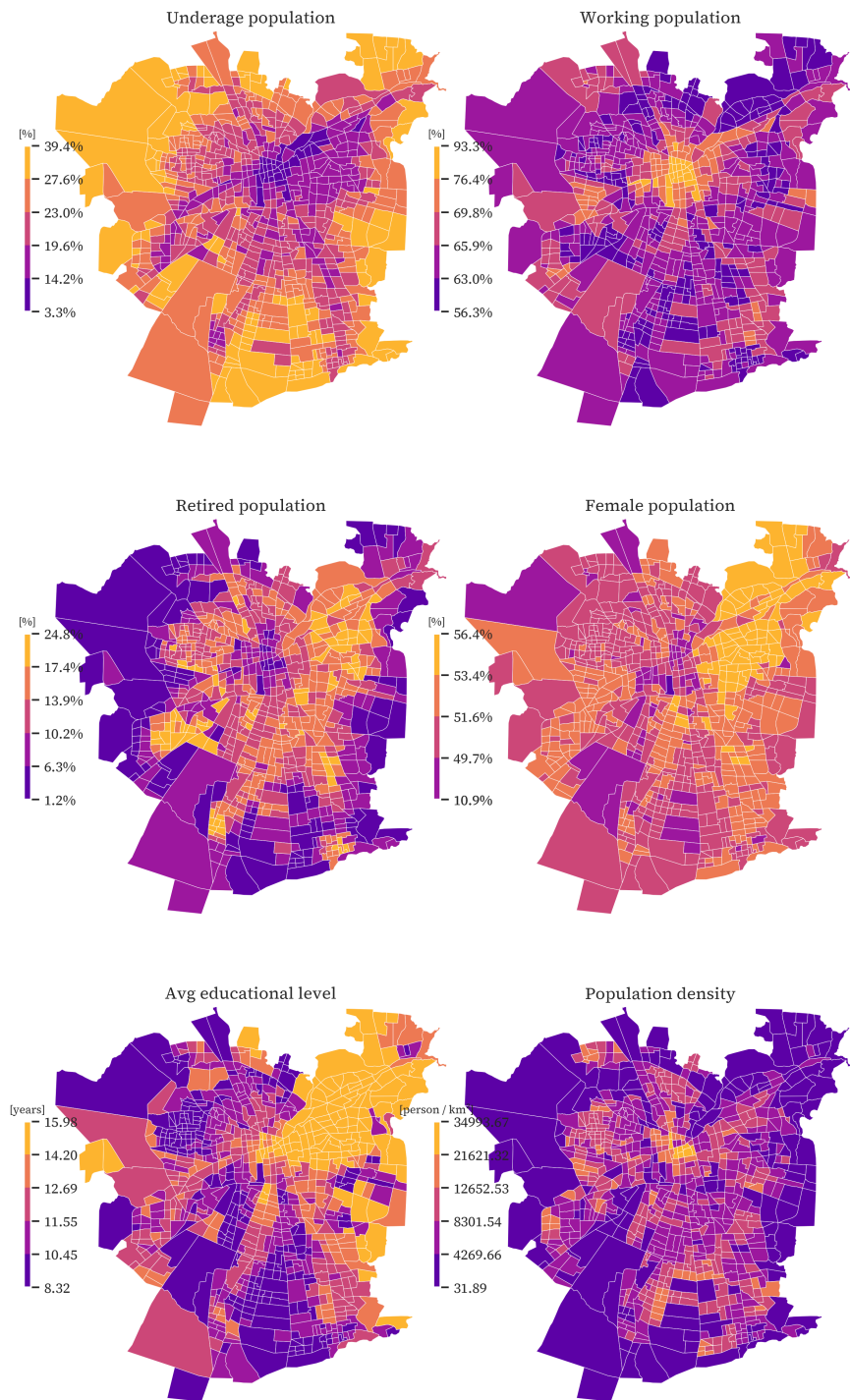


Figure 5: Spatial Distribution of Demographic Characteristics. This set of maps illustrates the spatial distribution of demographic indicators, including underage population, working population, retired population, female population, average educational level, and population density. These visualizations provide insights into the sociodemographic structure.



Figure 6: Spatial Distribution of Transportation and Mobility Metrics. This figure displays the spatial distribution of transportation infrastructure and usage metrics, such as the number of bus stops, bus stop density, bus stops per 100 people, average trips per person, average time of the first trip, and average trip distance.

of retired population, average educational level, percentage of female population, bus stop density, population density, average number of public transport trips made per person, average time of the first public transport trip of the day, and average public transport trip distance.

We implemented two clustering algorithms using Scikit-learn, a free, open-source Python library that provides tools for machine learning applications: k-means (Lloyd, 1982) and agglomerative hierarchical clustering. For agglomerative hierarchical clustering, we used the Ward method as the linkage criterion (Ward Jr, 1963), which minimizes the total within-cluster variance when merging clusters. We employed two different connectivity approaches for this method: a k-nearest neighbors (KNN) contiguity matrix and a sparse connectivity matrix. Each approach was applied across a range of 3 to 10 clusters.

Afterward, we used the Calinski-Harabasz score (Caliński and Harabasz, 1974) to compare the performance of these techniques and configurations to determine the best algorithm. The Calinski-Harabasz (CH) index, also known as the variance ratio criterion, measures the ratio of separation (the distance between cluster centroids) to cohesion (the distance between points within a cluster and their centroid). A higher score indicates better-defined clusters. As illustrated in Figure 7, the k-means algorithm with 3 clusters achieved the highest score, making it our chosen model.

The resulting regionalization (groups of zones) after applying the k-means clustering algorithm with $k=3$ is presented in Figure 8. The three clusters,

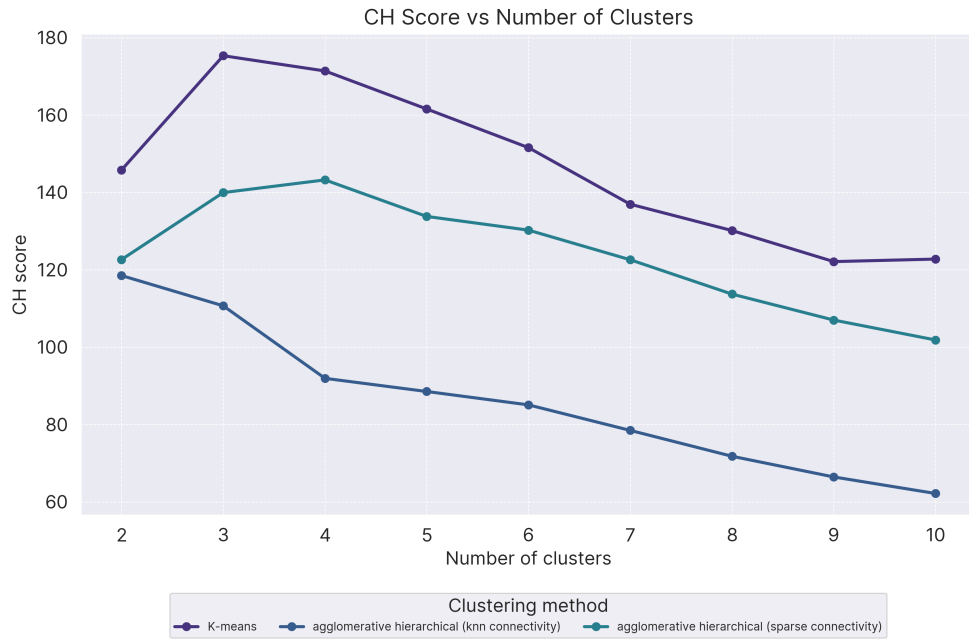


Figure 7: This figure shows the Calinski-Harabasz (CH) scores versus the number of clusters for three clustering methods: k-means (*purple*), agglomerative hierarchical with KNN connectivity (*blue*), and agglomerative hierarchical with sparse connectivity (*teal*). The x-axis represents the number of clusters (2 to 10), and the y-axis shows the CH score.

named according to their location in the city, almost form concentric circles over Santiago. The zones in light yellow represent the central area (*Central Region*), the zones in purple represent the periphery (*Peripheral Region*), and the zones in turquoise correspond to an intermediate area of the city (*Intermediate Region*).

After using the k-means algorithm to divide the city into three regions, we measured how the study variables were spread out in each region. This analysis revealed distinct characteristics for each region. Figure 9 and Figure 10 illustrate the characteristics of three distinct regions (*Central*, *Peripheral*, and *Intermediate*) in terms of demographic and transportation metrics. The *Central Region*, represented in *blue*, shows lower values in population categories (*underage*, *working*, and *female populations*) but higher *educational levels* and *bus stop density*. This region is densely populated and has a mostly adult population that is highly educated. Public transport trips originating in these areas are short, averaging less than a kilometer, and infrequent, with people making less than 2 trips per day on average. The *Peripheral Region*, in *yellow*, has a higher proportion of retired and female populations, alongside the highest population density and average trip distance. It also has the highest proportion of underage population and the lowest educational attainment among adults. Public transport trips starting in this zone are longer and more frequent than in the other two regions. The *Intermediate Region*, in *purple*, balances these metrics, displaying moderate values in all categories, suggesting a transitional nature between the *Central* and

Distribution of Zones (k-means clustering with $k=3$)

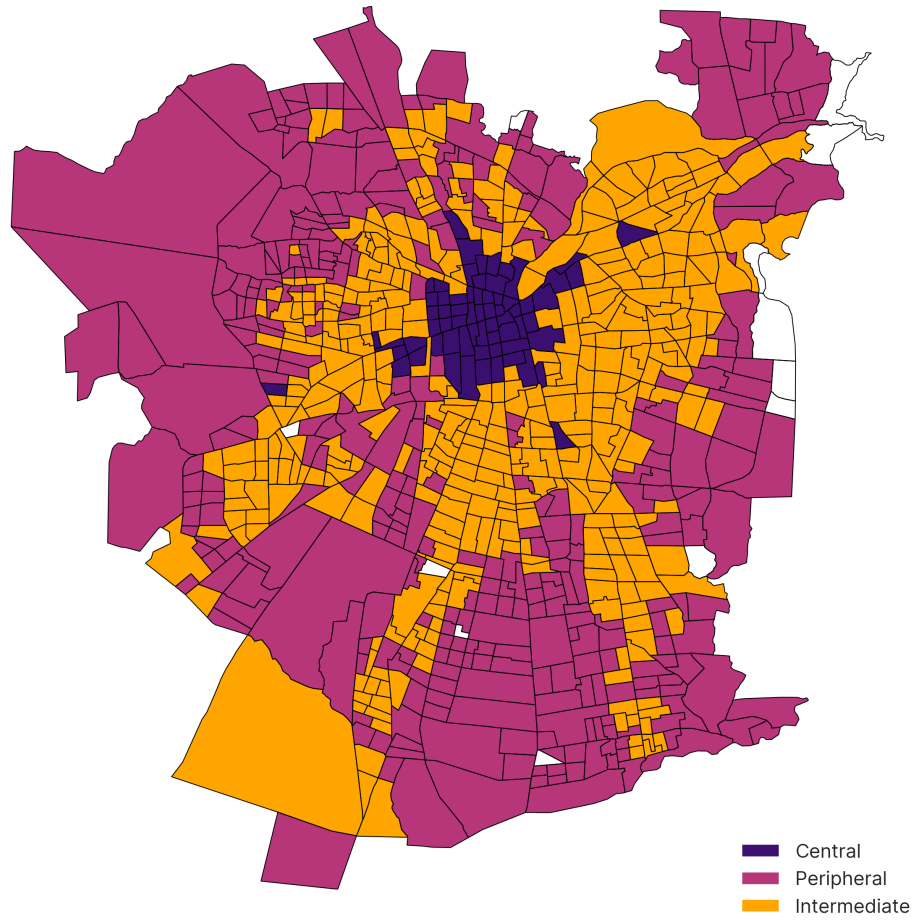


Figure 8: Spatial distribution of zones resulting from the k-means clustering algorithm with $k=3$. The zones in *blue* represent the *Central Region*, the zones in *orange* correspond to an intermediate area of the city (*Intermediate Region*), and the zones in *purple* represent the *Peripheral Region*.

Peripheral regions. Each cluster’s transportation usage is distinct, with the central region showing the highest average number of trips per person and the shortest average trip distance. It is important to note that the density of bus stops, and therefore the level of accessibility to public transport, is higher in the central zone than in the intermediate and peripheral zones.

After applying the clustering algorithm and dividing the zones into three groups of regions, we excluded 56 zones from the study. Although these zones had no origin and destination trips involving scooters, they were close to the treatment zones and therefore could still experience indirect mobility impacts post-intervention, potentially contaminating our control group. The criteria for exclusion were zones intersecting with a 1,440 m buffer applied to each scooter trajectory, a distance chosen based on the average trip distance calculated from our data. Figure 11 displays the control zones, treatment zones, and the excluded zones. Figure 12, along with Table 2, presents the distribution and detailed breakdown of control and treatment zones by cluster, excluding the 56 mentioned zones.

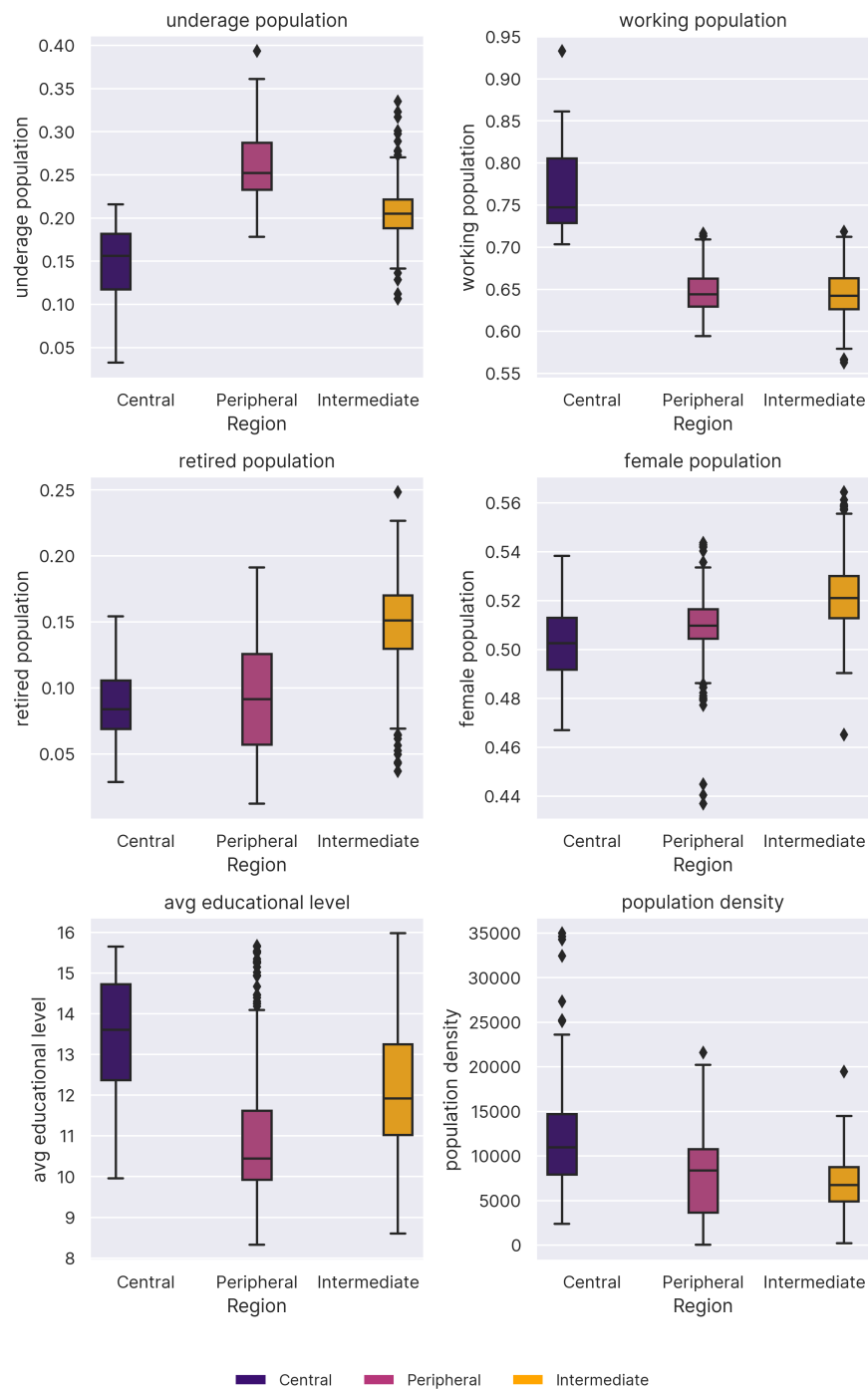


Figure 9: Demographic Characteristics Across Central, Peripheral, and Intermediate Regions. The figure shows box plots of demographic metrics for three regions: *Central* (blue), *Peripheral* (purple), and *Intermediate* (orange).

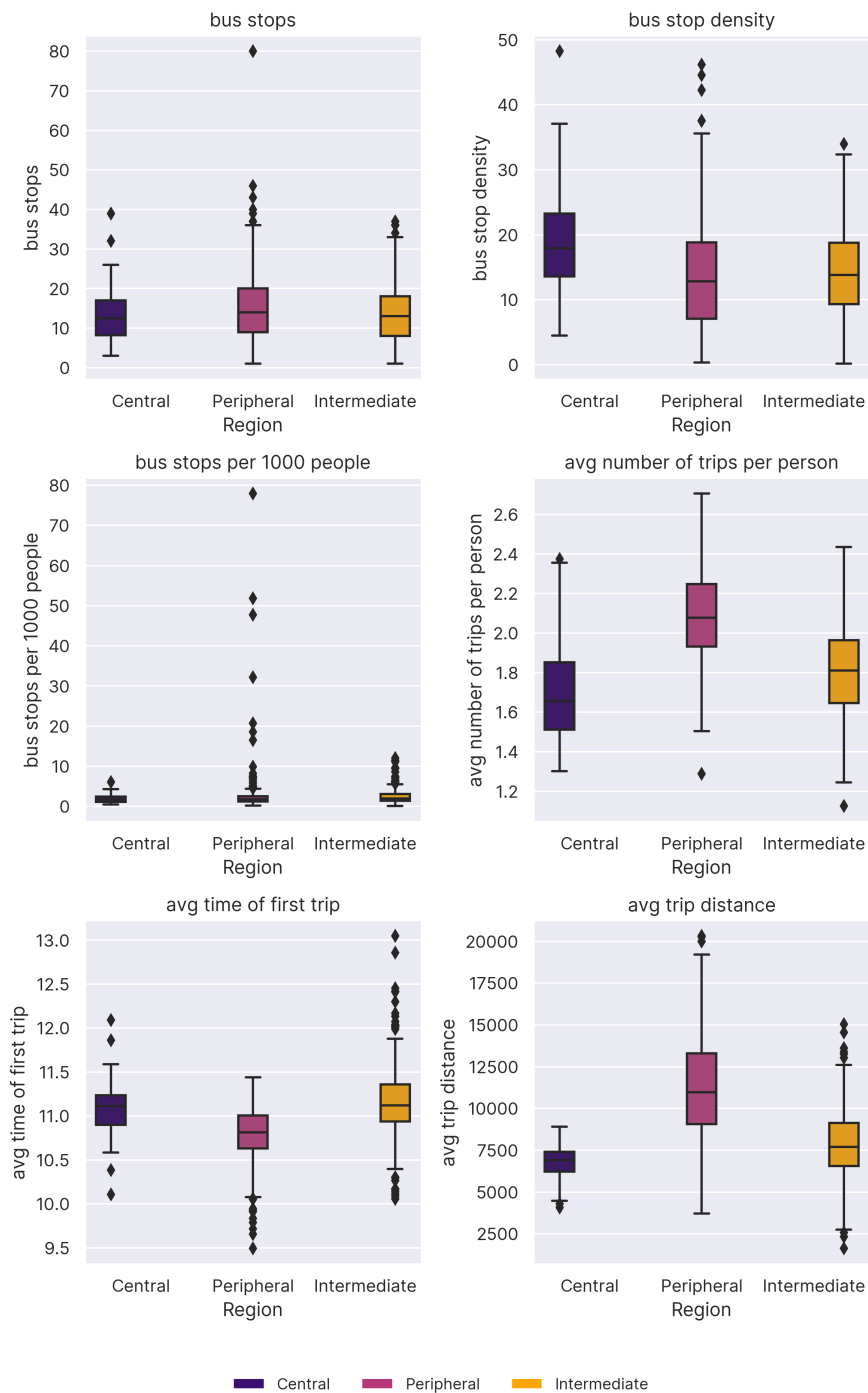


Figure 10: Transportation Characteristics Across Central, Peripheral, and Intermediate Regions. The figure shows box plots of transportation metrics for three regions: *Central* (blue), *Peripheral* (orange), and *Intermediate* (green).

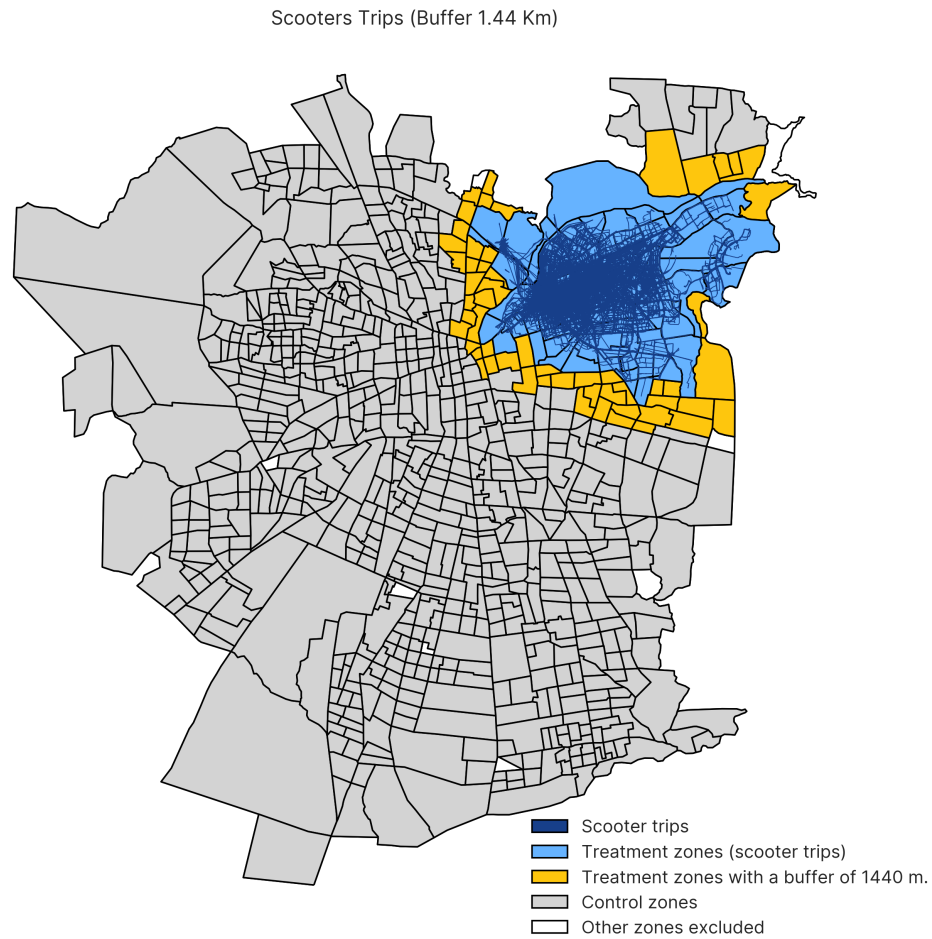


Figure 11: Spatial Distribution of Scooter Trips, Treatment, and Control Zones. The *dark blue* lines represent scooter trips. Treatment zones, highlighted in *light blue*, are those where scooter trips originated or ended. Zones highlighted in *yellow* are treatment zones with a buffer of 1,440 meters, chosen based on the average trip distance calculated from our data. Control zones, shown in *gray*, are not directly impacted by scooter trips or their buffer zones.

Distribution of Treatment and Control Zones by Region

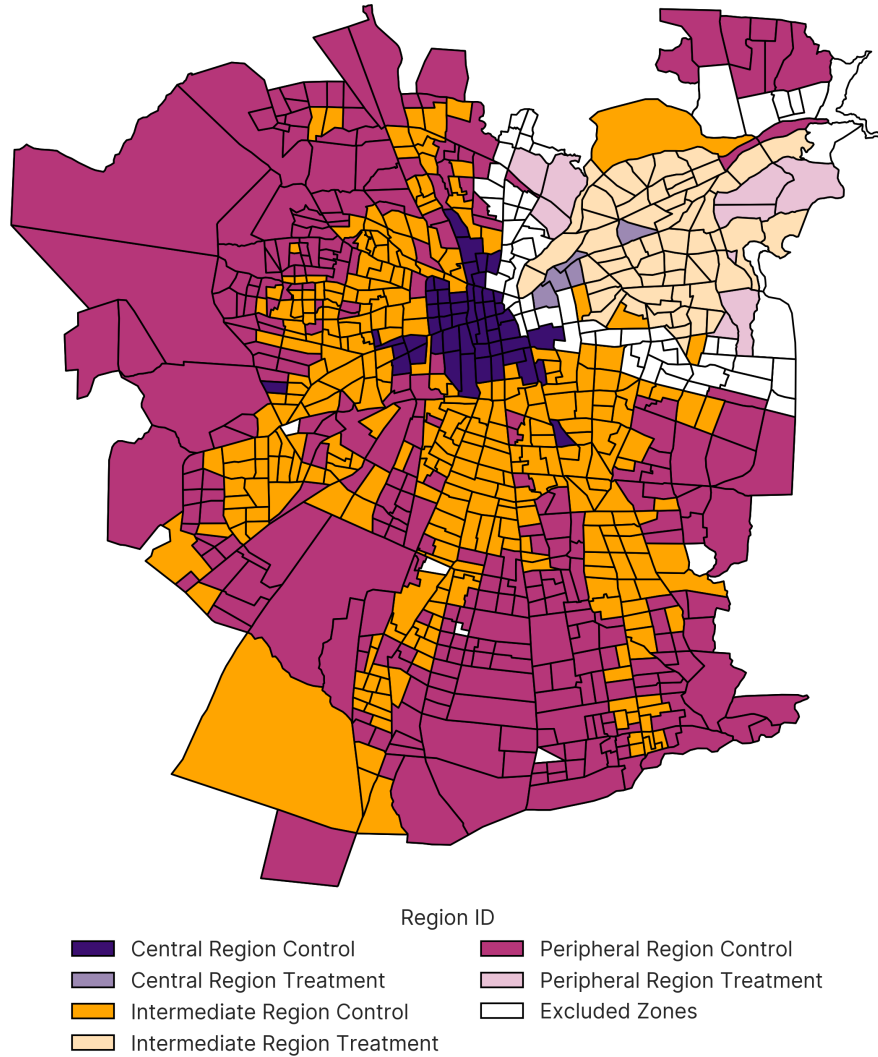


Figure 12: Distribution of the treatment and control zones by region. The map displays the zones within the study area. The figure presents the spatial distribution of zones across central, intermediate, and peripheral regions, categorized as either treatment or control zones. Darker shades correspond to control areas, while lighter shades denote treatment areas, with white zones representing regions excluded from the analysis

Region	Control Zones	Treatment Zones
Central Region	53	5
Intermediate Region	304	57
Peripheral Region	312	9

Table 2: Number of Control and Treatment Zones by Region. The column *Control Zones* indicates the number of control zones, while *Treatment Zones* indicates the number of treatment zones, excluding the 56 zones that intersect with a 1,440 m buffer around scooter trajectories.

Regression Model

Finally, to measure how the implementation of e-scooters affects public transportation demand, we applied a (DiD) approach in conjunction with two distinct Negative Binomial (NB) regression models (Cameron and Trivedi, 1986). The first model evaluates trips generation, focusing on trips originating from public transport stations, while the second evaluates trips attraction, analyzing trips ending at public transport stations. These models were chosen because public transportation trips often exhibit significant variability and over-dispersion, where the variance exceeds the mean, making traditional Poisson regression unsuitable for our dataset. The NB models effectively address over-dispersed count data, allowing us to better capture the variability in public transportation trips across different zones and providing a more accurate understanding of the factors influencing both trip generation and attraction scenarios.

The choice of NB models aligns with previous studies that have employed

similar approaches to analyze transportation demand across various modes (e.g., Schimohr and Scheiner, 2021; Hu et al., 2022; Manville et al., 2023). It is also consistent with prior research examining behavioral and spatial patterns in Santiago (Graells-Garrido et al., 2017; Beiró et al., 2018; Graells-Garrido et al., 2023b).

On the other hand, the DiD methodology estimates causal effects by comparing changes in outcomes over time between treatment and control groups before and after an intervention. In this study, the treatment group comprises Traffic Analysis Zones (TAZs) where shared e-scooter trips were observed, while the control group includes zones without e-scooter trips.

The DiD design assumes parallel trends: in the absence of shared e-scooter implementation, the treatment and control zones would exhibit similar trends in public transport demand. This assumption allows us to isolate the impact of the introduction of e-scooter from other factors that vary over time. The following equations represent the DiD structure incorporated into our NB regression models. Specifically, we employ two models: one focused on trip generation, and another on trip attraction. These models enable a comprehensive understanding of the effects of e-scooters on public transport demand.

The distinction between treatment and control zones, as detailed in the section Clustering and Definition of Treatment and Control Groups, is important to ensure the validity of the difference-in-differences (DiD) approach. Treatment zones were defined as those where shared e-scooter trips occurred,

while control zones were selected based on their lack of e-scooter trips and their location outside a 1,440-meter buffer zone. This approach minimizes the likelihood of indirect influences from nearby e-scooter activity on control zones. By applying this distinction within the regional clusters defined earlier, we ensured that comparisons between treatment and control zones reflect differences attributable to e-scooter introduction, rather than variations in sociodemographic or transport characteristics.

For the trip generation model, we examine how e-scooters arriving at a zone (destinations) affect the generation of public transport trips from that same zone, evaluating whether e-scooters function as feeders that bring people to transit stations. Conversely, for the trip attraction model, we analyze how e-scooters departing from a zone (origins) influence public transport trips ending in that zone, assessing whether e-scooters serve as last-mile connections from transit stations.

$$\begin{aligned}\ln(\mathbf{E}[y_{gen} \mid \mathbf{x}_{gen}]) &= \beta_0 + \beta_1 \text{treatment_zones} + \beta_2 \text{post} \\ &+ \beta_3(\text{treatment_zones} \times \text{post}) \\ &+ \beta \cdot \text{Controls} + \epsilon,\end{aligned}\tag{1}$$

$$\begin{aligned}\ln(\mathbf{E}[y_{att} \mid \mathbf{x}_{att}]) &= \beta_0 + \beta_1 \text{treatment_zones} + \beta_2 \text{post} \\ &+ \beta_3(\text{treatment_zones} \times \text{post}) \\ &+ \beta \cdot \text{Controls} + \epsilon,\end{aligned}\tag{2}$$

where y_{gen} and y_{att} represent the count of public transport trips in a TAZ

for trip generation and attraction scenarios, respectively. The variable *treatment_zones* is a dummy variable indicating whether the zone is associated with e-scooter trips: for the trip generation model, it takes the value 1 if e-scooter trips end in the zone (destinations), and for the trip attraction model, it takes the value 1 if e-scooter trips start in the zone (origins). The variable *post* is a dummy indicating the post-implementation period (2019), and the interaction term *treatment* $\times$ *post* captures the difference-in-differences (DiD) effect.

The $\beta \cdot \text{Controls}$ term incorporates additional control variables to account for variations in trip counts due to socio-demographic, temporal, and spatial factors. These control variables include the type of day (Monday-to-Thursday, Friday, Saturday, or Sunday, with Monday-to-Thursday as the reference category) and the time period during which trips occur (morning valley, morning peak, lunch, afternoon valley, afternoon peak, night, or night valley, with morning valley as the reference category). Spatial and demographic characteristics such as the log-transformed count of the working population in the TAZ, the average educational level relative to the regional average, and the number of metro lines within a 750-meter radius are also included. By accounting for these variables, the models ensure robustness in estimating the effects of e-scooter implementation, isolating its impact from other potential confounding factors.

For each scenario (generation and attraction), we estimated three separate models: (1) a model focusing on trips whose first or last stage involves a bus;

(2) a model focusing on trips whose first or last stage involves the metro; and (3) a model that includes all public transport trips that involve either bus or metro in any stage of the journey. In all models, the dependent variable is the count of public transport trips in each Traffic Analysis Zone (TAZ). We applied the regression to three regions (*Central*, *Intermediate*, and *Peripheral*), each containing its respective control and treatment areas.

Our analysis focused on two distinct time periods in fall: for public transport, we used data on boardings and alightings from the second week of May in both 2018 and 2019; for scooters, we calculated the average number of trips between March 1 and April 15, 2019. Ideally, we would have used data for both modes from the same week—either in April or May—but this was not possible due to data availability. We therefore assumed that e-scooter usage remained relatively stable from the March–April period to mid-May.

A difference-in-differences approach was utilized to control for temporal trends and other unobservable differences between the zones that implemented scooters (treatment group) and those that did not (control group).

This approach captures the effects within each region individually, providing a more detailed understanding of the relationship between scooter implementation and public transportation usage across different zones and transport modes. The inclusion of specific days of the week and periods of the day in our models accounts for variations in transportation usage patterns due to different daily and temporal dynamics.

For the difference-in-differences design, we established a minimum thresh-

old of 5 daily e-scooter trips per zone as the treatment assignment criterion. This threshold ensures that only zones with sufficient e-scooter activity are considered "treated" by the intervention. Based on empirical analysis of zone-day observations for both e-scooter origins (403 observations) and destinations (420 observations), the 5-trip threshold is positioned between the 25th percentile (3.0 trips/day for both) and the median (8.0 trips/day for origins, 8.75 trips/day for destinations). This results in 64.3% of origin observations and 66.9% of destination observations being assigned to treatment, with consistent assignment rates across flow types (2.6 percentage point difference). The threshold effectively excludes zones with sporadic activity that would not be meaningfully exposed to e-scooter interventions, preserving the internal validity of the difference-in-differences design.

5. Results

In this section, we present our findings on the impact of shared e-scooter introduction on public transportation usage in Santiago, Chile. Using negative binomial regression with a difference-in-differences approach, we analyze how the deployment of e-scooter services affects public transport demand across urban regions. Our analysis distinguishes between trip generation (public transport boardings - trips originating in each zone) and attraction (public transport alightings - trips ending in each zone) to capture the multidirectional effects of this new micro-mobility service.

Generation of Trips

Tables in Appendix C present the results of the negative binomial models for public transport trip generation. Table 3 displays the marginal effects of the introduction of shared e-scooters on public transport trip generation across different regions. Table 4 shows the average public transport trips by region and mode of transport, providing baseline context for interpreting the percentage changes in trip generation following e-scooter introduction.

We examine how e-scooter trips arriving at each zone influence the generation of public transport trips from those same zones. This analysis reveals whether e-scooters function as feeders to public transport (bringing people to transit stations who then continue their journeys via bus or metro) or whether they operate independently. Our findings demonstrate spatially heterogeneous effects across regions and transport modes.

In the Central Region, the introduction of shared e-scooters is associated with a statistically significant reduction in total public transport boardings of 465.81 trips per hour per zone, representing a 23.87% decrease. When examining individual transport modes separately, neither bus boardings (20.93 additional trips per hour per zone, 11.01% increase) nor metro boardings (450.92 additional trips per hour per zone, 10.17% increase) show statistically significant changes.

In the Intermediate Region, the introduction of shared e-scooters shows contrasting effects across transport modes. Total public transport boardings increased significantly by 178.99 trips per hour per zone, corresponding to a

33.64% increase. Metro boardings alone also experienced a statistically significant increase of 210.53 trips per hour per zone (9.77% increase). However, bus boardings showed a minimal and non-significant decrease of 0.53 trips per hour per zone (-0.59%).

In the Peripheral Region, no statistically significant effects were observed across any transport mode. Metro boardings could not be evaluated for this region due to the limited number of metro stations.

Mode of Transport	Central	Intermediate	Peripheral
Bus and Metro	-465.81* (-23.87%*)	178.99* (33.64%*)	52.43 (18.23%)
Bus	20.93 (11.01%)	-0.53 (-0.59%)	4.30 (4.99%)
Metro	450.92 (10.17%)	210.53* (9.77%*)	– –

Table 3: Difference-in-differences estimates of e-scooter impact on public transport trip generation (boardings). Marginal effects are expressed as additional trips per hour per zone, with percentage changes relative to baseline levels (Table 4) shown in parentheses. Estimates are derived from negative binomial regression models controlling for temporal, spatial, and demographic factors. Results in bold with asterisks indicate statistical significance at the 5% level ($p < 0.05$).

Region	Boardings (trips/hr/zone)			Alightings (trips/hr/zone)		
	Bus	Metro	Bus or Metro	Bus	Metro	Bus or Metro
Central	190.09	4435.24	1951.40	134.21	1186.10	540.08
Intermediate	89.26	2155.23	531.86	62.78	460.68	136.52
Peripheral	86.04	–	287.60	49.97	–	71.12

Table 4: Baseline public transport demand by region and mode. Values represent average trips per hour per zone during the pre-treatment period (May 2018), providing context for interpreting percentage changes in Tables 3 and 5. Boardings capture trip origins (generation model) while alightings capture trip destinations (attraction model).

Attraction of Trips

Tables in Appendix D present the results of difference-in-differences models for public transport trip attraction, while Table 5 shows the marginal effects of e-scooter introduction on public transport trips ending in different urban regions. The table includes both absolute values (additional trips per hour per zone) and percentage changes in parentheses, allowing for a comprehensive assessment of the impact across all transport modes and regions.

Mode of Transport	Central	Intermediate	Peripheral
Bus and Metro	-15.58 (-2.88%)	5.57* (4.08%*)	3.07 (4.32%)
Bus	13.88 (10.34%)	1.31 (2.09%)	4.25 (8.51%)
Metro	9.19 (0.77%)	-35.52 (-7.71%)	– –

Table 5: Difference-in-differences estimates of e-scooter impact on public transport trip attraction (alightings). Values represent marginal effects in additional trips per hour per zone, with percentage changes relative to baseline levels (Table 4) in parentheses. Results from negative binomial regression models controlling for temporal, spatial, and demographic factors. Values in bold and with asterisks are statistically significant at $p < 0.05$.

Regarding the impact of e-scooter introduction on public transport alightings, the analysis reveals heterogeneous effects across regions and transport modes. For total public transport alightings (bus and metro), the Intermediate Region showed the only statistically significant effect with an increase of 5.57 trips per hour per zone (4.08%), while the Central Region showed a non-significant decrease (-15.58 trips per hour per zone, -2.88%) and the Peripheral Region showed a non-significant increase (3.07 trips per hour per

zone, 4.32%). For individual transport modes, all effects were non-significant. Bus alightings demonstrated increases across all regions (Central: 13.88 trips per hour per zone, 10.34%; Intermediate: 1.31 trips per hour per zone, 2.09%; Peripheral: 4.25 trips per hour per zone, 8.51%), while metro alightings exhibited mixed patterns with an increase in the Central Region (9.19 trips per hour per zone, 0.77%) and a decrease in the Intermediate Region (-35.52 trips per hour per zone, -7.71%). Metro alightings could not be evaluated in the Peripheral Region due to limited metro infrastructure coverage.

5.1. Robustness and Model Validation

We examine the robustness of our difference-in-differences estimates by comparing our baseline specification with an extended model that includes demographic controls. This analysis serves two purposes: first, it assesses the stability of treatment effects when controlling for observable characteristics that may correlate with both treatment assignment and travel outcomes; second, it provides insights into the mechanisms through which the intervention affects travel behavior across different urban areas.

Our baseline specification (Model 1) includes the core DiD variables (treatment zones, post-period indicator, and their interaction), *metro* accessibility, and *type of day* and *time period* fixed effects. The extended specification (Model 2) augments this baseline with additional demographic controls: *log working population* and *relative educational level*. We examine these specifications across three distinct geographic areas that capture different urban

contexts: Central Region, Intermediate Region, and Peripheral Region.

Tables 6 and 7 present the results for trip generation and attraction models, respectively, across the three geographic regions: Central, Intermediate, and Peripheral. Both specifications include metro accessibility as a fundamental control for public transit infrastructure. Model 2 extends the baseline by adding demographic controls (log working population and relative educational level) to assess the robustness of treatment effects to observable characteristics that may correlate with both treatment assignment and travel outcomes.

For trip generation, treatment effects remain statistically significant and economically meaningful in both the baseline model (without sociodemographic controls) and the extended model (with controls) for central and intermediate areas. In central areas, the negative treatment effect decreases modestly from -0.282 to -0.249 when controls are introduced, but retains significance at the 5% level. This suggests successful modal substitution or trip consolidation in areas with high existing accessibility. In intermediate areas, the positive and significant effect declines from 0.413 to 0.358, indicating induced demand or improved connectivity. In contrast, the effect observed in the peripheral region is sensitive to the inclusion of sociodemographic controls: the coefficient decreases from 0.224 to 0.185 and loses statistical significance, suggesting that socioeconomic characteristics play a key mediating role in these more remote locations.

The robustness analysis of total public transportation usage (metro and

Variable	Central	Intermediate	Peripheral
<i>Model 1: Baseline DiD + Metro</i>			
Treatment $\times$ Post	-0.2818* (0.055)	0.4134* (0.094)	0.2241* (0.088)
Metro	0.657* (0.142)	0.075 (0.055)	-0.027 (0.079)
R-squared	0.024	0.031	0.031
<i>Model 2: DiD + Metro + Demographic Controls</i>			
Treatment $\times$ Post	-0.2491* (0.052)	0.3583* (0.089)	0.1854 (0.091)
Metro	0.229* (0.041)	0.178* (0.033)	0.055 (0.052)
Log working population	0.352* (0.065)	0.356* (0.078)	0.198* (0.089)
Relative education level	0.688* (0.112)	0.374* (0.095)	0.249* (0.108)
R-squared	0.036	0.037	0.034

Table 6: Robustness analysis for public transport trip generation models across geographic regions. Values represent regression coefficients from negative binomial models, with standard errors clustered at the zone level in parentheses. Model 1 includes core difference-in-differences variables, metro accessibility, and temporal fixed effects. Model 2 adds demographic controls (log working population and relative educational level). Values in bold with asterisks are statistically significant at $p < 0.05$.

bus boardings and alightings) reveals several important findings. First, the inclusion of demographic controls consistently improves model fit across regions and specifications, as evidenced by increases in R-squared values. This improvement indicates that demographic characteristics capture meaningful variation in travel behavior that would otherwise remain in the error term.

Second, the comparison between trip generation and attraction models reveals a coherent asymmetry that reflects e-scooter adoption patterns, particularly in the intermediate region. Treatment effects are substantially larger

Variable	Central	Intermediate	Peripheral
<i>Model 1: Baseline DiD + Metro</i>			
Treatment $\times$ Post	0.0054 (0.043)	0.0276 (0.039)	0.0486 (0.067)
Metro access	0.530* (0.089)	0.049 (0.041)	0.115 (0.067)
R-squared	0.029	0.025	0.024
<i>Model 2: DiD + Metro + Demographic Controls</i>			
Treatment $\times$ Post	-0.0306 (0.041)	0.0418* (0.037)	0.0436 (0.065)
Metro access	0.132* (0.035)	0.103* (0.028)	0.179* (0.045)
Log working population	0.308* (0.058)	0.281* (0.062)	0.075* (0.071)
Relative education level	0.630* (0.098)	0.341* (0.085)	0.168* (0.095)
R-squared	0.039	0.030	0.025

Table 7: Robustness analysis for public transport trip attraction models across geographic regions. Values represent regression coefficients from negative binomial models, with standard errors clustered at the zone level in parentheses. Model 1 includes core difference-in-differences variables, metro accessibility, and temporal fixed effects. Model 2 adds demographic controls (log working population and relative educational level). Values in bold with asterisks are statistically significant at $p < 0.05$.

and more robust for trip generation than for trip attraction, consistent with e-scooters functioning primarily as first-mile connectivity solutions in intermediate areas, where users predominantly employ these devices to access public transportation nodes rather than for last-mile distribution from transit stops to final destinations. This contrasts with the central region, where e-scooters appear to substitute for public transportation in trip generation, while showing no significant impact on trip attraction patterns.

Third, the persistence of significant treatment effects in the presence of

demographic controls demonstrates the robustness of our findings. The fact that estimated effects remain statistically significant and of substantial magnitude after controlling for observable characteristics that could correlate with treatment assignment strengthens the causal interpretation of our results. The modest changes in coefficient magnitudes suggest that our treatment effects are not merely reflecting unobserved heterogeneity in demographic composition across treatment and control areas.

The consistency of our results across specifications and geographic areas provides strong evidence for the robustness of our causal estimates. Treatment effects maintain their magnitude and significance when including demographic controls, while the distinct spatial patterns observed across central and intermediate areas validate that we are capturing genuine behavioral responses to e-scooter availability rather than confounding factors.

6. Discussion

Our results suggest both dynamics can coexist within the same city but in different zones: substitution dominates where abundant transit offers redundant options, while complementarity emerges where e-scooters address genuine accessibility gaps. These findings underscore the importance of tailoring micromobility policy to local transit conditions rather than assuming uniform impacts across urban areas.

Our findings reveal that shared e-scooter impacts on public transport are fundamentally shaped by urban context, transport mode, and local accessi-

bility conditions. Rather than uniform substitution or complementarity, we observe spatially differentiated patterns that account for the varied findings in existing literature.

Spatial Heterogeneity Explains Mixed Literature Findings

The coexistence of substitution effects in central areas (-23.87% in total boardings) and complementarity in intermediate zones (+33.64% in boardings) within the same urban system resolves the apparent contradiction in previous studies. Research reporting predominantly substitutive effects (Lu et al., 2024; Laa and Leth, 2020) may have focused on high-density areas similar to our Central Region, while studies emphasizing complementarity (Bai and Jiao, 2020; Vinagre Díaz et al., 2023) may have examined areas resembling our Intermediate Region. However, the significant combined bus-metro effect without corresponding individual mode significance in central areas suggests caution in interpreting underlying mechanisms.

This spatial variation aligns with findings from Oslo (Aarhaug et al., 2023) and Indianapolis (Luo et al., 2021), where both relationships occurred simultaneously within the same city. Our contribution lies in systematically demonstrating how built environment characteristics (transit density, trip distances, and accessibility levels) predict which relationship dominates in different urban areas.

Mode-Specific Effects Reflect Service Characteristics

Metro services show stronger complementary relationships with e-scooters than bus services across all regions. The significant increase in metro boardings in the Intermediate Region (+9.77%) contrasts with neutral bus effects, suggesting that fixed-route rail services benefit more from first/last-mile connections than flexible bus networks. This pattern aligns with Berlin findings where e-scooter demand increased near new metro stations (Weschke, 2023).

Bus services' consistent neutrality across regions may reflect Santiago's extensive, high-frequency network characteristics that make them less susceptible to e-scooter substitution. The comprehensive coverage and frequent stops of integrated bus systems may provide sufficient flexibility to maintain ridership even as alternative short-distance options emerge.

Urban Structure Drives Access Patterns

The strongest effects occur in the Intermediate Region, characterized by moderate transit density and intermediate trip distances. These conditions create optimal environments for e-scooters to bridge accessibility gaps without competing directly with efficient transit services. Central areas show substitution effects in combined bus and metro trips (-23.87%), though individual mode effects are not statistically significant, potentially reflecting statistical power limitations when disaggregating data or effects occurring primarily in multimodal trips. Peripheral areas show no significant effects, likely because trip distances exceed practical e-scooter range.

Policy Implications

These patterns suggest spatially differentiated strategies rather than uniform policies. Central areas could benefit from dynamic pricing that encourages e-scooter use during peak congestion periods while protecting transit during off-peak hours. Intermediate areas should focus on integration through coordinated infrastructure and payment systems to strengthen the observed complementary effects. Peripheral areas may require targeted deployment strategies to address mobility gaps.

The apparent substitution effects observed in central areas could potentially benefit system efficiency if they reduce peak-hour transit congestion, though this interpretation requires further investigation given the statistical patterns observed. Rather than restricting e-scooters, policies might consider strategic approaches to optimize mobility systems, while recognizing the need for additional research to understand underlying mechanisms.

Limitations and Future Research

Our analysis operates at the zonal level due to data integration constraints, potentially masking individual-level heterogeneity. The focus on Santiago limits generalizability, though the systematic spatial approach provides a framework for examining other contexts. Short-term effects may not capture longer-term adaptation patterns as users optimize multimodal journeys.

Future research should examine temporal variations to understand whether

substitution effects help manage peak-hour congestion or create capacity challenges. Integration of individual-level data across transport modes would enable more precise understanding of modal shift dynamics and support development of adaptive policies that respond to real-time demand patterns across different urban contexts.

7. Conclusions

This study demonstrates spatially heterogeneous effects of shared e-scooter introduction on public transport usage in Santiago, Chile. Central areas experienced substitution effects (23.87% reduction in total boardings), intermediate areas showed complementary effects with increases in trip generation (33.64%) and attraction (4.08%), while peripheral areas exhibited no significant impacts.

These spatially differentiated patterns help explain the divergent findings in existing literature and underscore the need for context-sensitive policies that recognize e-scooters' variable roles across urban environments. Metro services demonstrated stronger complementary relationships with e-scooters than bus services, particularly for first/last-mile connections. Our automated clustering methodology provides a replicable framework for future quasi-experimental studies, contributing to more rigorous comparative research across different urban contexts.

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Appendix A. Time blocks

Time	Period
06:01 - 09:00:59	Morning Peak
09:01 - 12:00:59	Morning Valley
12:01 - 14:00:59	Lunch
14:01 - 17:30:59	Afternoon Valley
17:31 - 20:30:59	Afternoon Peak
20:31 - 23:00:59	Night Valley
23:01 - 06:00:59	Night

Table A.8: Time blocks used in the study

Appendix B. Supplementary Summary of Zone Clustering Variables

Name	min	max
under age population	3.25	39.35
working population	56.28	93.30
retired population	1.22	24.84
female population	10.90	56.44
avg. educational level	8.32	15.98
bus stops	0.00	80.00
bus stop density	0.00	48.25
bus stops per 1000 people	0.00	77.93
population density	31.88	34993.67
avg. number of trips per person	1.00	2.71
avg. time of first trip	9.49	13.05
avg. trip distance	1627.34	20322.69

Table B.9: **Summary of Zone Clustering Variables.** This table provides the minimum and maximum values for each variable used in our study.

Appendix C. Results of Negative Binomial Models for Evaluating Generation of Public Transport Trips

Parameters	Peripheral Region	Intermediate Region	Central Region
Intercept	4.3777 (71.393) ***	3.1707 (40.445) ***	3.7286 (15.994) ***
Time periods			
Morning peak	0.5620 (25.938) ***	0.2413 (10.778) ***	-0.0879 (-1.548)
Lunch	-0.5106 (-23.628) ***	-0.3235 (-14.510) ***	-0.1018 (-1.798) *
Afternoon peak	-0.0617 (-2.848) **	0.3198 (14.506) ***	0.5771 (10.277) ***
Afternoon valley	0.0017 (0.079)	0.2867 (13.044) ***	0.4843 (8.656) ***
Night	-1.8128 (-81.101) ***	-2.3724 (-105.902) ***	-2.3969 (-41.747) ***
Night valley	-1.3019 (-59.505) ***	-0.6664 (-29.799) ***	-0.4283 (-7.574) ***
Type of day			
Friday	0.0714 (4.163) ***	0.0561 (3.223) **	0.0550 (1.233)
Saturday	-0.4290 (-24.587) ***	-0.4753 (-26.956) ***	-0.5434 (-12.099) ***
Sunday	-0.8734 (-48.838) ***	-0.9535 (-52.715) ***	-1.1174 (-24.561) ***
Difference-in-differences variables			
Post	0.0117 (0.992)	-0.0520 (-3.942) ***	-0.0114 (-0.358)
Treatment_zones	-0.3488 (-4.707) ***	-0.2635 (-8.019) ***	-0.4151 (-4.306) ***
Treatment_zones x post	0.1854 (1.873) *	0.3583 (11.194) ***	-0.2491 (-2.265) **
Infrastructure			
Metro	0.0549 (4.364) ***	0.1777 (13.239) ***	0.2289 (4.188) ***
Sociodemographic information			
Log(working population)	0.1977 (28.755) ***	0.3562 (38.206) ***	0.3524 (13.045) ***
Relative educational level	0.2488 (29.738) ***	0.3741 (36.668) ***	0.6883 (30.801) ***
Model performance			
Nº observations	36,368	44,975	6,724
Log-likelihood	-2.3001e+05	-3.0176e+05	-5.3445e+04

*p < 0.10; **p < 0.05; ***p < 0.01. Z-values are shown in parenthesis.

Table C.10: Results of negative binomial models for evaluating generation of bus and metro trips.

Parameters	Peripheral Region	Intermediate Region	Central Region
Time periods			
Morning peak	158.9475 (24.731) ***	120.5449 (10.701) ***	-164.4222 (-1.547)
Lunch	-144.4176 (-22.934) ***	-161.6223 (-14.338) ***	-190.3694 (-1.796) *
Afternoon peak	-17.4446 (-2.846) **	159.7837 (14.330) ***	1079.1796 (9.799) ***
Afternoon valley	0.4828 (0.079)	143.2316 (12.897) ***	905.5947 (8.365) ***
Night	-512.7257 (-64.484) ***	-1185.1764 (-76.167) ***	-4482.0023 (-30.069) ***
Night valley	-368.2161 (-51.614) ***	-332.9000 (-28.389) ***	-800.8503 (-7.406) ***
Type of day			
Friday	20.1870 (4.157) ***	28.0295 (3.220) **	102.7591 (1.232)
Saturday	-121.3347 (-23.900) ***	-237.4285 (-26.039) ***	-1016.1518 (-11.555) ***
Sunday	-247.0365 (-44.296) ***	-476.3533 (-46.916) ***	-2089.4423 (-20.936) ***
Difference-in-differences variables			
Post	3.2993 (0.992)	-25.9553 (-3.938) ***	-21.2911 (-0.357)
Treatment zones	-98.6635 (-4.700) ***	-131.6356 (-7.976) ***	-776.2562 (-4.282) ***
Treatment zones $\times$ post	52.4331 (1.873) *	178.9864 (11.069) ***	-465.8094 (-2.262) **
Infrastructure			
Metro	15.5300 (4.362) ***	88.7940 (13.095) ***	428.1174 (4.156) ***
Sociodemographic information			
Log(working population)	55.9234 (27.747) ***	177.9331 (36.064) ***	658.8805 (12.011) ***
Relative educational level	70.3743 (28.174) ***	186.9059 (33.840) ***	1287.0477 (23.736) ***

*p < 0.10; **p < 0.05; ***p < 0.01. Z-values are shown in parentheses.

Table C.11: Marginal effects of negative binomial models for evaluating generation of metro and bus trips by region.

Parameters	Peripheral Region	Intermediate Region	Central Region
Intercept	2.3816 (67.622) ***	1.5371 (35.311) ***	2.9284 (24.234) ***
Time periods			
Morning peak	0.5526 (45.303) ***	0.2293 (19.913) ***	0.0924 (2.081) **
Lunch	-0.1522 (-11.824) ***	0.0150 (1.190)	0.1693 (5.386) ***
Afternoon peak	-0.1416 (-11.863) ***	0.1739 (15.604) ***	0.4935 (17.313) ***
Afternoon valley	-0.1180 (-9.971) ***	0.1737 (15.623) ***	0.4439 (15.586) ***
Night	-1.5104 (-116.104) ***	-1.9181 (-162.780) ***	-1.9647 (-71.540) ***
Night valley	-1.1450 (-88.931) ***	-0.5830 (-48.824) ***	-0.4331 (-14.516) ***
Type of day			
Friday	0.0346 (3.561) ***	0.0307 (3.381) ***	0.0332 (1.482)
Saturday	-0.4784 (-47.678) ***	-0.4809 (-51.631) ***	-0.4736 (-20.614) ***
Sunday	-0.8771 (-83.860) ***	-0.9001 (-93.325) ***	-0.9397 (-39.669) ***
Difference-in-differences variables			
Post	-0.0204 (-3.025) **	-0.0387 (-5.752) ***	-0.1795 (-11.306) ***
Treatment_zones	0.1019 (2.577) **	-0.1688 (-10.853) ***	0.4402 (8.800) ***
Treatment_zones x post	0.0511 (0.937)	-0.0060 (-0.339)	0.1121 (1.689) *
Infrastructure			
Metro	0.1433 (19.865) ***	-0.0547 (-7.984) ***	0.4810 (16.558) ***
Sociodemographic information			
Log(working population)	0.2742 (70.098) ***	0.3806 (72.965) ***	0.2040 (14.851) ***
Relative educational level	0.1038 (24.288) ***	0.2075 (42.176) ***	0.2777 (23.306) ***
Model performance			
N ^o observations	102,881	118,438	21,486
Log-likelihood	-5.3523e+05	-6.1880e+05	-1.2555e+05

*p < 0.10; **p < 0.05; ***p < 0.01. Z-values are shown in parenthesis.

Table C.12: Results of negative binomial models for evaluating generation of bus trips.

Parameters	Peripheral Region	Intermediate Region	Central Region
Time periods			
Morning peak	46.5049 (43.318) ***	20.1732 (19.789) ***	-0.4427 (-0.081)
Lunch	-13.3951 (-11.796) ***	1.3160 (1.190)	31.6000 (5.368) ***
Afternoon peak	-11.9123 (-11.831) ***	15.3047 (15.546) ***	92.0866 (16.749) ***
Afternoon valley	-9.9317 (-9.953) ***	15.2853 (15.560) ***	82.8260 (15.162) ***
Night	-127.1050 (-98.592) ***	-168.7648 (-129.826) ***	-366.6060 (-54.728) ***
Night valley	-96.3604 (-80.193) ***	-51.2927 (-47.290) ***	-80.8218 (-14.244) ***
Type of day			
Friday	2.9136 (3.560) ***	2.7022 (3.380) ***	6.2007 (1.482)
Saturday	-40.2563 (-46.023) ***	-42.3135 (-49.867) ***	-88.3664 (-19.964) ***
Sunday	-73.8129 (-76.349) ***	-79.1927 (-84.575) ***	-175.3454 (-35.764) ***
Difference-in-differences variables			
Post	-1.7151 (-3.024) **	-3.4088 (-5.749) ***	-33.5015 (-11.165) ***
Treatment zones	8.5729 (2.577) **	-14.8500 (-10.827) ***	82.1445 (8.674) ***
Treatment zones x post	4.2967 (0.937)	-0.5301 (-0.339)	20.9254 (1.688) *
Infrastructure			
Metro	12.0582 (19.742) ***	-4.8133 (-7.972) ***	89.7584 (16.228) ***
Sociodemographic information			
Log(working population)	23.0765 (65.909) ***	33.4870 (67.721) ***	38.0689 (14.385) ***
Relative educational level	8.7374 (24.047) ***	18.2544 (40.908) ***	51.8139 (22.186) ***

*p < 0.10; **p < 0.05; ***p < 0.01. Z-values are shown in parentheses.

Table C.13: Marginal effects of negative binomial models for evaluating generation of bus trips by region.

Parameters	Intermediate Region	Central Region
Intercept	8.5065 (57.937) ***	6.2008 (26.294) ***
Time periods		
Morning peak	0.2523 (6.568) ***	-0.1863 (-3.845) ***
Lunch	-0.3441 (-8.987) ***	-0.0803 (-1.607)
Afternoon peak	0.4737 (12.547) ***	0.6747 (13.975) ***
Afternoon valley	0.3013 (8.039) ***	0.5419 (11.254) ***
Night	-3.4498 (-90.987) ***	-3.4877 (-70.918) ***
Night valley	-0.5930 (-15.565) ***	-0.3613 (-7.497) ***
Type of day		
Friday	0.0563 (2.253) **	0.0552 (1.448)
Saturday	-0.6095 (-20.330) ***	-0.7370 (-19.158) ***
Sunday	-1.1318 (-35.975) ***	-1.4706 (-37.483) ***
Difference-in-differences variables		
Post	0.0274 (1.161)	-0.0488 (-1.840) *
Treatment_zones	0.4337 (7.949) ***	-0.2887 (-3.155) **
Treatment_zones x post	0.1012 (1.996) **	0.1024 (0.811)
Sociodemographic information		
Log(working population)	-0.1169 (-6.742) ***	0.2159 (7.932) ***
Relative educational level	0.1901 (10.186) ***	0.4170 (20.720) ***
Model performance		
N ^o observations	6,635	2,242
Log-likelihood	-5.3314e+04	-1.9238e+04

*p < 0.10; **p < 0.05; ***p < 0.01. Z-values are shown in parenthesis.

Table C.14: Results of negative binomial models for evaluating generation of metro trips.

Parameters	Intermediate Region	Central Region
Time periods		
Morning peak	524.6563 (6.521) ***	-820.6119 (-3.823) ***
Lunch	-715.6997 (-8.867) ***	-353.5639 (-1.666) *
Afternoon peak	985.0912 (12.214) ***	2971.7075 (13.117) ***
Afternoon valley	626.5734 (7.945) ***	2386.4923 (10.795) ***
Night	-7174.3981 (-54.013) ***	-15360.0000 (-41.987) ***
Night valley	-1233.1786 (-14.965) ***	-1591.3766 (-7.378) ***
Type of day		
Friday	137.8575 (2.250) **	243.0036 (1.446)
Saturday	-1267.5468 (-19.168) ***	-3246.0942 (-17.468) ***
Sunday	-2353.7217 (-30.772) ***	-6476.8729 (-28.807) ***
Difference-in-differences variables		
Post	56.8927 (1.161)	-215.1262 (-1.837) *
Treatment destination zones	901.9012 (7.843) ***	-1271.6845 (-3.142) ***
Treatment zones $\times$ post	210.5348 (1.994) **	450.9161 (0.811)
Sociodemographic information		
Log(working population)	-243.0255 (-6.675) ***	950.9396 (7.768) ***
Relative educational level	395.3644 (10.053) ***	1836.4571 (18.737) ***

*p < 0.10; **p < 0.05; ***p < 0.01. Z-values are shown in parentheses.

Table C.15: Marginal effects of negative binomial models for evaluating generation of metro trips by region.

Appendix D. Results of Negative Binomial Models for Evaluating Attraction of Public Transport Trips

Parameters	Peripheral Region	Intermediate Region	Central Region
Intercept	3.3678 (88.758) ***	2.5196 (54.925) ***	3.2381 (22.265) ***
Time periods			
Morning peak	0.2828 (21.819) ***	0.1537 (11.730) ***	0.1627 (4.822) ***
Lunch	0.2202 (16.270) ***	0.0098 (0.711)	-0.1602 (-4.620) ***
Afternoon peak	0.9948 (79.825) ***	0.3999 (31.433) ***	-0.0851 (-2.575) **
Afternoon valley	0.6297 (51.501) ***	0.2616 (20.915) ***	-0.0434 (-1.331)
Night	-1.4215 (-100.573) ***	-2.2359 (-167.240) ***	-3.0218 (-96.218) ***
Night valley	0.3266 (24.384) ***	-0.3893 (-28.602) ***	-1.0225 (-28.996) ***
Type of day			
Friday	0.0237 (2.339) **	0.0383 (3.757) ***	0.0430 (1.669) *
Saturday	-0.4823 (-45.604) ***	-0.4905 (-46.614) ***	-0.5355 (-20.258) ***
Sunday	-0.9193 (-83.384) ***	-0.9338 (-85.465) ***	-1.0701 (-39.063) ***
Difference-in-differences variables			
Post	0.0098 (1.384)	-0.0099 (-1.306)	0.0378 (2.070) **
Treatment_zones	-0.0785 (-1.521)	-0.1131 (-6.183) ***	-0.4432 (-7.702) ***
Treatment_zones x post	0.0436 (0.613)	0.0418 (2.048) **	-0.0306 (-0.390)
Infrastructure			
Metro	0.1795 (23.085) ***	0.1028 (13.630) ***	0.1316 (4.007) ***
Sociodemographic information			
Log(working population)	0.0752 (17.429) ***	0.2806 (50.788) ***	0.3079 (18.051) ***
Relative educational level	0.1681 (33.372) ***	0.3412 (60.611) ***	0.6298 (49.526) ***
Model performance			
Nº observations	130,832	152,443	27,018
Log-likelihood	-6.5106e+05	-8.2614e+05	-1.7272e+05

*p < 0.10; **p < 0.05; ***p < 0.01. Z-values are shown in parenthesis.

Table D.16: Results of negative binomial models for evaluating attraction of bus and metro trips.

Parameters	Peripheral Region	Intermediate Region	Central Region
Time periods			
Morning peak	19.9543 (21.684) ***	20.4836 (11.718) ***	82.9149 (4.809) ***
Lunch	15.5340 (16.190) ***	1.3097 (0.711)	-84.7186 (-4.607) ***
Afternoon peak	70.1871 (70.766) ***	53.2886 (30.724) ***	-43.3861 (-2.572) **
Afternoon valley	44.4302 (49.047) ***	34.8545 (20.698) ***	-22.1223 (-1.331)
Night	-100.2964 (-89.925) ***	-297.9469 (-125.879) ***	-1540.2396 (-58.082) ***
Night valley	23.0469 (24.123) ***	-51.8750 (-28.229) ***	-521.1965 (-26.443) ***
Type of day			
Friday	1.6727 (2.339) **	5.0994 (3.756) ***	21.9143 (1.668) *
Saturday	-34.0297 (-44.065) ***	-65.3573 (-44.930) ***	-272.9721 (-19.368) ***
Sunday	-64.8611 (-75.347) ***	-124.4369 (-76.670) ***	-545.4409 (-34.068) ***
Difference-in-differences variables			
Post	0.6898 (1.384)	-1.3164 (-1.306)	19.2547 (2.069) **
Treatment zones	-5.5352 (-1.521)	-15.0703 (-6.178) ***	-225.8979 (-7.642) ***
Treatment zones $\times$ post	3.0729 (0.613)	5.5681 (2.048) **	-15.5797 (-0.390)
Infrastructure			
Metro	12.6630 (22.830) ***	13.7034 (13.570) ***	67.0650 (3.996) ***
Sociodemographic information			
Log(working population)	5.3062 (17.402) ***	37.3967 (49.008) ***	156.9238 (17.158) ***
Relative educational level	11.8625 (32.499) ***	45.4711 (56.300) ***	320.9919 (39.655) ***

*p < 0.10; **p < 0.05; ***p < 0.01. Z-values are shown in parentheses.

Table D.17: Marginal effects of negative binomial models for evaluating attraction of metro and bus trips by region.

Parameters	Peripheral Region	Intermediate Region	Central Region
Intercept	1.5522 (46.917) ***	1.3196 (33.973) ***	3.4830 (31.749) ***
Time periods			
Morning peak	0.2968 (25.964) ***	0.1404 (13.320) ***	0.1534 (5.780) ***
Lunch	0.1959 (16.421) ***	-0.0103 (-0.922)	-0.1692 (-5.949) ***
Afternoon peak	0.8740 (79.589) ***	0.2905 (28.351) ***	-0.2110 (-8.112) ***
Afternoon valley	0.5680 (52.686) ***	0.2135 (21.153) ***	-0.0448 (-1.746) *
Night	-1.2645 (-100.594) ***	-1.8244 (-167.301) ***	-2.2068 (-88.479) ***
Night valley	0.2430 (20.555) ***	-0.4401 (-40.027) ***	-1.1807 (-42.503) ***
Type of day			
Friday	0.0081 (0.908)	0.0271 (3.291) ***	0.0333 (1.640)
Saturday	-0.4765 (-51.119) ***	-0.4536 (-53.056) ***	-0.4341 (-20.785) ***
Sunday	-0.8750 (-89.882) ***	-0.8604 (-97.142) ***	-0.8876 (-40.977) ***
Difference-in-differences variables			
Post	-0.0358 (-5.777) ***	-0.0564 (-9.221) ***	-0.1808 (-12.515) ***
Treatment_zones	0.5419 (12.060) ***	-0.0773 (-5.462) ***	-0.1088 (-2.373) **
Treatment_zones x post	0.0867 (1.396)	0.0211 (1.276)	0.1060 (1.721) *
Infrastructure			
Metro	0.1459 (21.716) ***	-0.0249 (-4.103) ***	0.4183 (15.775) ***
Sociodemographic information			
Log(working population)	0.2484 (66.134) ***	0.3563 (75.823) ***	0.1387 (11.094) ***
Relative educational level	0.0358 (8.836) ***	0.2135 (48.373) ***	0.3555 (33.653) ***
Model performance			
N ^o observations	130,168	151,127	26,685
Log-likelihood	-6.1321e+05	-7.4018e+05	-1.4671e+05

*p < 0.10; **p < 0.05; ***p < 0.01. Z-values are shown in parenthesis.

Table D.18: Results of negative binomial models for evaluating attraction of bus trips.

Parameters	Peripheral Region	Intermediate Region	Central Region
Time periods			
Morning peak	14.5498 (25.757) ***	8.7350 (13.299) ***	20.0847 (5.763) ***
Lunch	9.6039 (16.359) ***	-0.6415 (-0.922)	-22.1579 (-5.931) ***
Afternoon peak	42.8414 (72.624) ***	18.0721 (28.025) ***	-27.6286 (-8.064) ***
Afternoon valley	27.8394 (50.678) ***	13.2786 (21.012) ***	-5.8653 (-1.745) *
Night	-61.9803 (-92.099) ***	-113.4862 (-136.316) ***	-289.0129 (-63.333) ***
Night valley	11.9088 (20.441) ***	-27.3759 (-39.330) ***	-154.6264 (-37.946) ***
Type of day			
Friday	0.3968 (0.908)	1.6837 (3.290) ***	4.3616 (1.640)
Saturday	-23.3565 (-49.445) ***	-28.2178 (-51.683) ***	-56.8494 (-20.223) ***
Sunday	-42.8876 (-82.027) ***	-53.5216 (-88.691) ***	-116.2411 (-37.348) ***
Difference-in-differences variables			
Post	-1.7549 (-5.767) ***	-3.5092 (-9.209) ***	-23.6747 (-12.353) ***
Treatment zones	26.5597 (12.023) ***	-4.8109 (-5.460) ***	-14.2478 (-2.372) **
Treatment zones x post	4.2511 (1.396)	1.3094 (1.276)	13.8759 (1.720) *
Infrastructure			
Metro	7.1517 (21.563) ***	-1.5514 (-4.101) ***	54.7859 (15.533) ***
Sociodemographic information			
Log(working population)	12.1759 (63.498) ***	22.1658 (71.042) ***	18.1703 (10.864) ***
Relative educational level	1.7545 (8.826) ***	13.2804 (46.836) ***	46.5557 (31.138) ***

*p < 0.10; **p < 0.05; ***p < 0.01. Z-values are shown in parentheses.

Table D.19: Marginal effects of negative binomial models for evaluating attraction of bus trips by region.

Parameters	Intermediate Region	Central Region
Intercept	7.4561 (68.157) ***	6.0602 (25.746) ***
Time periods		
Morning peak	0.2481 (8.567) ***	0.2224 (4.820) ***
Lunch	0.0360 (1.193)	-0.1746 (-3.590) ***
Afternoon peak	0.5623 (20.076) ***	-0.0421 (-0.939)
Afternoon valley	0.3622 (13.131) ***	-0.0265 (-0.595)
Night	-2.4647 (-75.216) ***	-3.0731 (-60.502) ***
Night valley	-0.3028 (-10.215) ***	-0.9904 (-20.778) ***
Type of day		
Friday	0.0587 (2.569) **	0.0435 (1.188)
Saturday	-0.5464 (-22.864) ***	-0.6368 (-16.616) ***
Sunday	-0.9734 (-38.500) ***	-1.1742 (-28.558) ***
Difference-in-differences variables		
Post	0.0225 (1.317)	0.0473 (1.821) *
Treatment_zones	0.7056 (17.541) ***	-0.2021 (-2.332) **
Treatment_zones x post	-0.0769 (-1.732) *	0.0079 (0.066)
Sociodemographic information		
Log(working population)	-0.1866 (-14.286) ***	0.0851 (3.116) ***
Relative educational level	0.2042 (15.812) ***	0.4994 (26.150) ***
Model performance		
N ^o observations	24,579	9,283
Log-likelihood	-1.6595e+05	-7.0499e+04

*p < 0.10; **p < 0.05; ***p < 0.01. Z-values are shown in parenthesis.

Table D.20: Results of negative binomial models for evaluating attraction of metro trips.

Parameters	Intermediate Region	Central Region
Time periods		
Morning peak	114.5955 (8.534) ***	257.9006 (4.791) ***
Lunch	16.6482 (1.192)	-202.4349 (-3.579) ***
Afternoon peak	259.7544 (19.091) ***	-48.8484 (-0.939)
Afternoon valley	167.3319 (12.848) ***	-30.7297 (-0.595)
Night	-1138.6288 (-55.833) ***	-3563.4397 (-39.943) ***
Night valley	-139.8859 (-10.162) ***	-1148.4773 (-19.108) ***
Type of day		
Friday	27.1079 (2.567) **	50.4060 (1.187)
Saturday	-252.4360 (-21.858) ***	-738.4166 (-15.640) ***
Sunday	-449.6835 (-34.309) ***	-1361.5697 (-24.684) ***
Difference-in-differences variables		
Post	10.3876 (1.317)	54.7903 (1.820) *
Treatment zones	325.9625 (16.743) ***	-234.3963 (-2.328) **
Treatment zones x post	-35.5166 (-1.732) *	9.1316 (0.066)
Sociodemographic information		
Log(working population)	-86.2191 (-13.930) ***	98.6420 (3.108) ***
Relative educational level	94.3309 (15.452) ***	579.0490 (22.690) ***

*p < 0.10; **p < 0.05; ***p < 0.01. Z-values are shown in parentheses.

Table D.21: Marginal effects of negative binomial models for evaluating attraction of metro trips by region.